



Robotics

PROJECT PROPOSAL

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I. SUMMARY	3
II. INTRODUCTION	4
III. NEEDS/PROBLEMS	5
IV. GOALS/OBJECTIVES	5
V. PROCEDURES/SCOPE OF WORK.....	5
VI. TIMETABLE	6
VII. BUDGET.....	6
XI. NEXT STEPS	6
XII. APPENDIX	7

1. Summary

The main purpose of the project is to fully automate Dobot Magician to play chess against a human player using a chess engine. The expected result of the project is to have a robotic arm with 4 degrees of freedom and a camera that detects the player's move on the chess board and perform the best move for the current situation. The total anticipated budget is 1000 EGP.

2. Introduction

The project aims to exemplify cutting-edge advancements in technology and robotics by showcasing the DOBOT robotic arm's ability to surpass the strategic prowess of an experienced human in the game of chess. The DOBOT, a highly versatile and programmable robotic system, serves as the central component of this project, offering four degrees of freedom to replicate human hand movements. Its sophisticated set of motors facilitates movement along four degrees of freedom.

A strategically positioned camera, mounted atop the robotic arm, captures the chessboard's current state. Upon the player's move, a designated switch triggers the camera to capture the chessboard image. Subsequently, a Python script processes the image, visually representing the player's move on a dedicated web application designed for the project. This information is then transmitted to a connected chess engine, which analyzes the current board state and determines the optimal move. The decision is promptly displayed on the web app, and the Python script issues instructions to the DOBOT to physically replicate the move on the chessboard.

The DOBOT actively engages with the player by providing feedback on their moves. Positive comments are articulated for commendable moves, while alerts are issued for incorrect ones, fostering a learning experience for the players. At the conclusion of the game, the robotic arm executes a symbolic handshake with the user, symbolizing mutual respect between players.

This innovative project is set to be featured at the university's chess tournament, providing an opportunity for participants to challenge the AI-powered DOBOT. The ultimate winner of the competition earns the privilege of facing off against the DOBOT, creating an exciting test of human-AI gaming capabilities. This project not only showcases technological prowess but also enhances the educational aspect of chess playing, creating a captivating and interactive experience for participants and spectators alike.

3. Needs/Problems

There are different kinds of problems to be faced throughout the project. Firstly, controlling the arm and its degrees of freedom. Secondly, applying inverse kinematics to be able to move the arm from one position to another. Lastly, the communication between the DOBOT, web app, and the chess engine in order to send and receive data to perform outputs.

4. Goals/Objectives

- Learn how to control the DOBOT and study all its restrictions to know how to make it move free so that it can play chess efficiently.
- Learn how to apply inverse kinematics on a robotic arm.
- Learn more about chess engines and the AI behind it.

5. Procedures/Scope of Work

The initial phase of the project involves the fabrication of 3D-printed chess pieces, setting the foundation for an engaging and visually appealing chessboard. Prior to diving into the intricacies of chess gameplay, a meticulous testing phase is planned for the DOBOT robotic arm. This preliminary stage focuses on validating the arm's basic functionality, such as gripping and releasing objects using a suction cup. By confining these test actions to a controlled environment, any potential errors or inefficiencies in the robotic arm can be identified and addressed before advancing to the chess-specific functionalities.

Following successful testing and refinement of the DOBOT's fundamental capabilities, the project progresses to the implementation of inverse kinematics. This crucial step allows precise control over the arm's movements, enabling it to navigate to specific cells on the chessboard. The objective is to ensure that the robotic arm seamlessly and accurately responds to positional commands, laying the groundwork for the intricate maneuvers required in chess gameplay.

Once the robotic arm is deemed ready, demonstrating proficiency in various movements, the chess engine and camera components are integrated into the system. This marks the commencement of the testing and training phase, where the DOBOT learns to play chess. Through iterative testing, the team refines the communication between all project devices, addressing any potential errors or challenges that arise during gameplay.

This systematic approach ensures a methodical progression from foundational testing to the sophisticated integration of chess-specific functionalities. By addressing potential issues at each stage, the project aims to achieve a robust and seamlessly functioning system that combines advanced robotics, chess logic, and interactive gameplay.

6. Timetable

	Description of Work	Start and End Dates
Phase One	Research Phase	19/03/2023 to 26/11/2023
Phase Two	Print chess pieces	26/11/2023 to 3/12/2023
Phase Three	Program the DOBOT to move using inverse kinematics	3/12/2023 to 12/04/2023
Phase Four	Program the camera to take input and translate it to the DOBOT	12/04/2023 to 21/04/2023
Phase Five	Integrate the DOBOT with the chess engine	21/04/2023 to 24/12/2023

Budget

	Component (Quantity)	Price
First	Integrate the robot with the chess engine	700 EGP
Second	Extras (jumpers, 4mm bolts and nuts, screws, domain, chess board, etc.)	300 EGP
	Total	1000 EGP

7. Next Steps

The future work to be done on the project.

- Build a database that keeps records of all the matches played by the robot against a specific user and the match difficulty.
- The robot can detect if the user has performed a wrong move. For now, the robot will react to that by giving an alert but in future the robot will also be able to put that piece back to its previous position if the move was wrong.
- Add voice control feature to enable people with disabilities to play against the arm.

8. Appendix

- <https://stockfishchess.org>
- <https://www.dobot-robots.com/products/education/magician.html>